

ECO 202 — Practice Exam 4 — Version A

Worked solutions

Fall 2026 | Classes 14–18

Equivalent exact expressions and valid alternative reasoning are acceptable. For practice study, attempt the paper first, then compare reasoning, close these solutions, and reconstruct any missed method independently.

Multiple-choice key

Question	Answer	Main idea
1	C	Power is rejection probability at the specified alternative.
2	A	A null-based test evaluates variance at the null proportion.
3	C	Higher confidence requires a larger critical value.
4	A	Training received is a post-assignment variable.

1. C equals $1 - (1 - 1/2)^2 = 3/4$. A is the probability of zero successes at that alternative; B is the probability of exactly one success; D is the rejection probability under the null, the Type I error probability $1 - (9/10)^2$. The rule need not have a conventional 5% size to have a well-defined power.
2. A uses the null variance. B is estimate-based and appropriate for the specified Wald interval instead. C is a variance, not its square root; D omits division by the sample size inside the square root.
3. C follows from a larger critical value at the fixed standard error. A and B contradict that width change. D confuses the confidence level with the point estimate.
4. A can be affected by the assignment itself and conditioning on it can change the total-effect target. B, C and D are measured before assignment and can be considered as prespecified adjustment variables, subject to a sound analysis plan; their being pre-treatment does not make every possible use automatically valid.

5. Changes within the same household

- (a) The target is μ_D , mean after-minus-before monthly spending change in the population sampled. Each household contributes one difference. Before/after measurements from the same household are generally dependent; the paired-difference standard deviation preserves that covariance.
- (b) $\widehat{SE} = 5/\sqrt{25} = 1$ dollar and $t = (2 - 0)/1 = 2$. Using the supplied rounded tail area, the two-sided p-value is approximately $2(0.0285) = 0.057$. Fail to reject at 5%; this does not prove the mean change is zero. Independent Normal differences give the exact t_{24} reference even if the two original spending columns are not separately Normal.
- (c) The interval is $2 \pm 2.064(1) = [-0.064, 4.064]$ dollars. Across repeated samples under the stated assumptions, this procedure covers the fixed mean change 95% of the time. It is not a 95% interval for individual household changes or a post-data probability that the fixed target lies in the reported endpoints. The before/after comparison alone does not isolate causation: other changes over time can affect spending.

- (d) Halving the standard error requires multiplying n by four, so the planning sample size is $4(25) = 100$ households. Indeed $\sqrt{25/100} = 1/2$. The question explicitly holds the critical value fixed; otherwise finite-sample t critical values would also change.

6. One employment proportion, two variance choices

- (a) $\hat{p} = 60/100 = 3/5$. Under $p_0 = 1/2$, the standard error is

$$\sqrt{\frac{(1/2)(1/2)}{100}} = \frac{1}{20}.$$

The statistic is $(3/5 - 1/2)/(1/20) = (1/10)/(1/20) = 2$.

- (b) The approximate two-sided p-value is $2(0.0228) = 0.0456$; reject at 5%. It is the null-reference probability of a statistic at least as extreme in either direction as the observed one, assuming the sampling and approximation conditions. It treats $p = 1/2$ as the premise, not a random event whose posterior probability is being calculated.

- (c) The estimate-based standard error is

$$\sqrt{\frac{(3/5)(2/5)}{100}} = \frac{\sqrt{6}}{50}.$$

The interval is

$$\left[\frac{3}{5} - 1.96 \frac{\sqrt{6}}{50}, \frac{3}{5} + 1.96 \frac{\sqrt{6}}{50} \right].$$

The null test uses the variance stipulated by $p_0 = 1/2$; the Wald interval estimates the variance with $\hat{p} = 3/5$. Hence exact algebraic interval-test equivalence is not guaranteed for this pair of procedures. Here both reject/exclude $1/2$: since $\sqrt{6} < 5/2$, the margin is smaller than $1.96(5/2)/50 = 0.098 < 0.1$.

- (d) Observed successes/failures are 60/40; null expected counts are 50/50. These support the respective Normal approximations. A voluntary poll can be selected in ways related to employment; large success/failure counts do not establish representativeness or independent probability sampling.

7. A complete joint distribution

- (a) The marginal probabilities are $3/20$, $9/20$ and $2/5$ for $X = 0, 1, 2$. Conditional on $X = 1$,

$$\mathbb{P}(Y = 0 \mid X = 1) = \frac{7/20}{9/20} = \frac{7}{9}, \quad \mathbb{P}(Y = 1 \mid X = 1) = \frac{1/10}{9/20} = \frac{2}{9}.$$

These sum to one.

- (b) Since Y is binary, its conditional mean is its conditional success probability:

$$f(0) = \frac{1/20}{3/20} = \frac{1}{3}, \quad f(1) = \frac{2}{9}, \quad f(2) = \frac{1/5}{2/5} = \frac{1}{2}.$$

The successive differences are $-1/9$ and $5/18$, not equal; f is not linear on the support.

- (c) Iterated expectations gives

$$\mathbb{E}[Y] = \frac{3}{20} \frac{1}{3} + \frac{9}{20} \frac{2}{9} + \frac{2}{5} \frac{1}{2} = \frac{1}{20} + \frac{1}{10} + \frac{1}{5} = \frac{7}{20}.$$

The marginal probability of $Y = 1$ is also $1/20 + 1/10 + 1/5 = 7/20$, and the mean of a binary variable is its success probability.

- (d) Within $X = 1$, the errors are $-2/9$ when $Y = 0$ and $7/9$ when $Y = 1$. Their weighted conditional mean is

$$\frac{7}{9} \left(-\frac{2}{9} \right) + \frac{2}{9} \left(\frac{7}{9} \right) = 0.$$

Mean-zero noise need not be zero for any one observation and does not require constant conditional variance or causality.

8. Group means and a regression prediction

- (a) The line is $\hat{y} = 5 + 3x$: the intercept is the noncollege group mean, 5 thousand dollars; the slope is the college-minus-noncollege difference, 3 thousand dollars of annual expenditure. With a binary indicator, a one-unit change moves from group 0 to group 1.

- (b) The standard error is

$$\sqrt{6^2/100 + 8^2/100} = \sqrt{(36 + 64)/100} = 1$$

thousand dollars. The 95% interval for the population mean difference is $3 \pm 1.96 = [1.04, 4.96]$ thousand dollars.

- (c) $H_0 : \mu_1 - \mu_0 = 0$ versus $H_a : \mu_1 - \mu_0 \neq 0$. The statistic is $z = 3/1 = 3$, the two-sided p-value is $2(0.00135) = 0.0027$, and equality is rejected at 5% under the independent-sample, large-sample reference.
- (d) The criticism is too strong: a stable association can support prediction without identifying an intervention effect. For future prediction, the relevant conditional expenditure distributions must remain sufficiently similar, with comparable measurement and population/support. The line predicts group means, not each individual's expenditure exactly. The data alone do not establish that sending a particular adult to college would increase that adult's expenditure by \$3,000, nor the population average causal effect of doing so; selection and confounding can account for the association.